

MD AL MOMEN

Software Engineering Student · Full-Stack Developer · Bioinformatics Researcher

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PROFESSIONAL SUMMARY

Software engineering student and full-stack developer with hands-on experience building and deploying production-grade web applications, alongside active research in bioinformatics and machine learning. Proficient in React, Node.js, Java, Python, and Firebase, with a strong record of taking projects from concept to live deployment. Passionate about applying deep learning and computer vision to solve real-world problems through creative, well-engineered solutions.

WORK EXPERIENCE

Assistant General Secretary

2025 – Present

Health Informatics Research Lab, Daffodil International University · Dhaka, Bangladesh

- Coordinate research activities and day-to-day lab operations in collaboration with the Principal Investigator.
- Support fellow researchers across bioinformatics and health informatics projects, including ML-based protein classification.
- Awarded Computer Vision Bootcamp completion certificate from the lab.

Full-Stack Web Developer

2025 – Present

GrihoTech — Home Appliances E-Commerce Platform · Remote

- Designed, built, and deployed a production-grade e-commerce platform using React 19, Vite, Node.js, Express, MySQL, and Firebase Authentication.
- Implemented an end-to-end pricing and discount system (original price, sale price, auto-calculated discount), multer-based image uploads, and Google Gemini AI product recommendations.
- Resolved complex Namecheap cPanel shared hosting deployment challenges (CloudLinux Node.js Selector, dist packaging, MySQL schema migrations).
- Integrated a React Admin dashboard with role-based access control and Firebase Storage CORS handling.

Researcher

Jun 2023 – Present

Health Informatics Research Lab, Daffodil International University · Dhaka, Bangladesh

- Conducting research on LLPS (Liquid-Liquid Phase Separation) protein classification using knowledge distillation with Mamba-based deep learning architectures.
- Developed and iterated KD pipelines (Response, Feature, Relation, Hybrid) achieving competitive benchmark results against the MambaPhase paper (Briefings in Bioinformatics, 2025).
- Supervisor: Dr. Md. Zahid Hasan, Associate Professor, CSE.

E-Commerce Web Developer

Jan 2024 – Present

absunifiedbd.com · Bangladesh

- Develop and maintain full-stack e-commerce features for a live commercial platform.
- Collaborate directly with business stakeholders to ship product management, order processing, and payment workflows.

SELECTED PROJECTS

GrihoTech — Home Appliances E-Commerce Platform2025 – Present

React 19 · Vite · Node.js · Express · MySQL · Firebase Auth · Google Gemini AI

- Full-stack e-commerce app with secure authentication, AI-powered product recommendations, dynamic pricing and discount engine, and admin dashboard. Deployed to production on Namecheap cPanel shared hosting.
- **Repository:** github.com/al-momen663/grihotech

Share Circle — Community Donation & Marketplace Platform2025

React 19 · TypeScript · Firebase · Firestore · Leaflet Maps · Netlify

- Donation and item-sharing platform supporting four user roles (Donor, Volunteer, Member, Admin) with real-time Firestore data, geo-mapped listings via Leaflet, and Firebase Cloud Functions backend.
- **Repository:** github.com/al-momen663/share-circle

BlitzMed — Emergency Ambulance Booking SystemJan 2025 – Apr 2025

HTML · CSS · JavaScript · Firebase

- Online platform enabling users to book ambulances in real time with live booking status and emergency request tracking.
- **Repository:** github.com/al-momen663/BlitzMed

DIU Transport SystemMay 2025 – Jun 2025

Web Application

- Online platform giving DIU students real-time access to university transport schedules and route information.
- **Repository:** github.com/al-momen663/DIU-transport-system

Medical Store Inventory AppJan 2025 – Apr 2025

Java · Desktop Application

- Inventory management system for medical stores to track stock levels, sales records, and supplier information.
- **Repository:** github.com/al-momen663/MedicalStoreApp

Lawyers Diary SystemJan 2025 – Present

Java · Desktop Application

- Client and case record management system for legal professionals, supporting case diaries, appointments, and client histories.

TECHNICAL SKILLS

Languages	Python, Java, C/C++, JavaScript (ES6+), TypeScript, HTML5, CSS3, SQL
Frontend	React 19, Vite, TypeScript, Tailwind CSS, Responsive Design
Backend	Node.js, Express.js, Spring Boot, REST APIs
Databases	MySQL, PostgreSQL, Firebase Firestore, SQLite
Cloud & DevOps	Firebase (Auth, Storage, Functions), cPanel, Netlify, Git, GitHub
ML & Research	PyTorch, Scikit-learn, Mamba architecture, Knowledge Distillation, Computer Vision
Tools	VS Code, IntelliJ IDEA, PyCharm, DataGrip, Postman, Linux

EDUCATION

B.Sc. in Software Engineering

Jan 2023 – Present

Daffodil International University, Dhaka, Bangladesh

GPA: 3.79 / 4.00

Higher Secondary Certificate (HSC)

Jul 2019 – Dec 2021

Chuadanga Government College, Bangladesh

GPA: 5.00 / 5.00 · Group: Science

Secondary School Certificate (SSC)

Jan 2018 – Jan 2019

Victoria Jubilee Govt. High School, Bangladesh

GPA: 5.00 / 5.00 · Group: Science

LEADERSHIP & ACTIVITIES

- **Volunteer**, DIU Voluntary Service Club (2023 – Present) — Active participant in community service and campus initiatives.
- **Active Donor**, DIU Blood Donor Club (2023 – Present) — Regular blood donor contributing to campus and community health programs.
- **General Member**, DIU Data Science Club (2023 – Present) — Participating in data science workshops, competitions, and knowledge-sharing sessions.

LANGUAGES

English (Professional working proficiency) · **Bangla** (Native)

REFERENCES

Dr. Md. Zahid Hasan

Associate Professor, Department of CSE, Daffodil International University

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Kaiser Ahmed

Principal Software Engineer, Oracle Cloud

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